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ASSESSMENT OF MODIFIED LEATHER WASTE ON CURE CHARACTERISTICS OF NATURAL RUBBER COMPOUND

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Abstract

Leather waste (buffing) were sourced, cleaned and modified. The modification revealed improved filler characteristics for both treated leather waste (TLW) and carbonized leather waste (CLW) in terms of, bulk density, moisture content, lignin content, cellulose content, hemicelluloses content, thermal stability and FTIR respectively. The results obtained were compared with carbon black (CB) filled vulcanizates. The cure characteristics of the compounded rubber showed improved scorch time (TLW; 2.59 - 2.02 min, CLW; 2.05 - 1.61 min, CB; 1.78 - 1.19 min) and cure time (TLW; 6.04 - 5.56 min, CLW; 5.93 - 4.51 min, CB; 4.77 - 4.45 min) decreased with filler loading while the cure rate index (TLW; 1.27 - 1.53 min^{-1}, CLW; 1.31 - 1.65 min^{-1}, CB; 1.40 - 1.74 min^{-1}), minimum torque (TLW; 4.10 - 5.08 kg-cm, CLW; 4.76 - 5.53 kg-cm, CB; 5.00 - 5.97 kg-cm) and maximum torque (TLW; 16.70 - 25.08 kg-cm, CLW; 19.44 - 26.19 kg-cm, CB; 20.04 - 27.95 kg-cm) increased with filler loadings from 10 phr – 50 phr. The research results showed that CLW and TLW can serve as alternative to CB for the production of polymer based articles such as shoe soles, floor-mats, oil seals, shock mounts etc. for some engineering applications.

Key Words: Buffing, Crosslink, FTIR, Leather, Vulcanizate

1. Introduction

The economic development of a country depends on the utilization of indigenous raw materials and their by-products (Li et al., 2017). As the pollution reduction can be possible by formulating cost intensive non-edible gelatin production thereby making leather industries eco-friendlier and environmentally compliant to buyers. Therefore, export earnings will be increased and the socio-economic status of people will be improved (El-Hout et al., 2022). Due to enhanced economic activities and rapid industrialization, tannery solid waste generation has increased dramatically in the last few decades. Tannery solid waste generation is a continually growing problem at regional and local levels due to rapid industrialization. Leather waste (buffing) is a fine powder of collagenous waste containing chromium, synthetic fat and other chemicals (Banon et al., 2021). Leather waste is generated when the leather is subjected to abrasion processes to achieve a uniform appearance. For every tone of skin or hide processed, about 2–6 kg of buffing dust is generated as solid waste. Circular economic strategies put pressure on the processing of wastes into secondary raw materials with the aim of their effective reuse instead of incineration or landfilling (Kumiawan et al., 2022). Recycling waste materials as filler for reinforcement of composites is of rising interest. Buffing dust is a tanning waste produced abundantly by local leather industry and its modification via chemical treatment and carbonization approach offers reduction of moisture absorption, increase in the surface roughness, decreasing the hydroxyl groups (Liu et al., 2019). Therefore, an attempt to use natural filler from leather waste as alternative filler for natural rubber was proposed (Popita et al., 2016). The choice of leather waste as the reinforcement in natural rubber composites is because it is renewable, biodegradable, non-toxic and abundantly available in Nigeria. It will also add value to local content thereby promoting industrial growth, economic development and sustainability.

1.1 Objective of Research Study

i. To obtain and characterize modified fillers
ii. To study the cure characteristics of the modified leather waste filled vulcanizates
iii. To assess the cure characteristics of the filled vulcanizates.

2.0 Materials and Method

2.1. Materials

The various materials used in this research include; Natural Rubber (NSR-10) obtained from Rubber Research Institute, Iyanomo, Benin City, Leather waste (Buffing) obtained from NILEST Metropolis and Environs, Zaria, Distilled water from Hadis Chemicals, Zaria, Kaduna State. Other materials obtained from British Drug House (BDH), England include; Sodium hydroxide (NaOH), Tetramethyl thiuram disulphide, Mercarpto benzothiazole sulphenamide, Stearic acid, Sulphur, Trimethylquinoline (TMQ), Zinc Oxide, Processing Oil (Paraffin wax), Carbon Black (N330), Methanol, Sodium Thiosulphate, Iodine Solution, Standard Starch Solution, Sulphonic Acid.

2.2. Method

2.2.1 Preparation and Modification of Leather Waste

Leather waste was collected from NILEST metropolis, Zaria, Kaduna State. One portion of the buffing was carbonized at 650°C while the other portion was washed thoroughly using water and dried in open air for a period of 120 hours prior to pulverizing. The pulverized buffing was further ball-milled and soaked in 20% solution of sodium hydroxide solution for 1h at room temperature. Then solution was decanted and the filler washed to attain neutrality after which the powder was dried at room temperature and oven-dried at 75^{0}C for 1 hour (Tenebe et al., 2021). The reaction equation is shown in equation 1

$$Filler - OH + NaOH \rightarrow Filler - ONa + H2O \quad 1$$

a

b

Plate I: Treated (a) and Carbonized Leather Waste (buffing) (b)

2.3 Characterization of Leather Waste (Buffing) Powder

The modified leather waste and carbon black-based fillers were characterized in terms of particle size, iodine value, moisture content, bulk density, pH, thermal stability, cellulose, hemicelluloses and lignin contents respectively and were used in compounding natural rubber.

2.3.1 Scanning Electron Microscope (SEM)

The micro-structural analysis was conducted in accordance with ASTM E2809-13. A specimen sample, usually non-conductive, was made conductive by coating with gold metal onto it and was cut into specified dimension using a sputter cutting machine (Li et al., 2020). The sample was placed on the column of the scanning electron microscope (SEM) where the image was focused using navigation camera and was transferred to electron mode in accordance to the desired magnification, which reveals the cracked areas, pores, bundles and voids present in the sample.

2.4 Characterization of the Natural Rubber

2.4.1 Determination of Dirt Content

The dirt content was determined in accordance with ASTM D1278-91a. The process involves homogenizing natural rubber in a two-roll mill and a test piece weighed out and cut into pieces. The piece of rubber is placed in a conical flask containing mineral turpentine. The flask with its content was placed in the oil bath and was heated to 135°C for 4 hours. On complete dissolution, the hot solution was filtered through a previously weighed, clean and dried sieve by decantation (Sharma et al., 2022). The dirt in the flask was washed into the sieve with a jet of warm mineral turpentine. The sieves with the dirt were dried in an oven and were recorded. The dirt content was calculated thus;

$$D(\%) = \frac{W_1}{W_2} \times 100 \quad 2$$

Where:

D = Dirt Content W_1= Weight of Dirt (g) W_2= Weight of test portion (g)

2.4.2 Determination of Volatile Matter

The volatile matter was determined in accordance with ASTM D5668-19. A test portion of 10g were weighed and passed twice through the Two-roll mill. The milled test piece was folded loosely and was placed on a properly labeled watch glass which was fed into an oven maintain at 100°C for 3h after which the test piece was removed and cooled for 30min. The volatile matter were expressed as percentage of the test portion (Tenebe et al., 2013).

$$Volatile\ Matter\ (\%) = \left(\frac{X - Z}{X}\right) \times 100 \quad 3$$

Where:

X = Weight of Test portion before drying Z = Weight of Test portion after drying

2.4.3 Determination of Ash Content

The ash content was determined in accordance with ASTM D5630. A test portion of 10g was weighed out from the homogenized piece and was wrapped in a filter paper. The wrapped test portion was placed in a covered crucible, which was previously ignited and weighed (Yang et al., 2017). The crucible with its contents was introduced into a muffle furnace controlled at a temperature of 550°C for 4h and was allowed to cool in desiccators, weighed to obtain the ash content as follows;

$$Ash\ (\%) = \left(\frac{X}{Z}\right)x100 \qquad 4$$

Where: X = weight of ash Z = weight of test portion

2.4.4 Determination of Plasticity Retention Index

The plasticity retention index was determined in accordance with ASTM D3194-17. The process involves cutting 20g test piece of the unaged (P_o) and aged (P_{30}) and then measuring the plasticity of rubber samples using a Wallace rapid plastometer (Zheng et al., 2020). The plasticity retention index was calculated using equation 5.

$$PRI\ (\%) = \left(\frac{P_{30}}{P_0}\right)x100 \qquad 5$$

Where:
P_{30} = aged sample P_o = unaged sample

2.5 Processing of Composites

Compounding of natural rubber with modified powdered filler was based on the formulation in Table 1

Table 1: Formulation in part per hundred rubber using batch factor

Ingredient	Parts per hundred rubber (phr)
Natural Rubber	100
Filler	Variable (10 - 50)
Zinc Oxide	5.0
Stearic acid	2.5
Sulphur	1.5
Mercarpto	1.5
benzothiazole	3.5
sulphenamide	5.0
Tetramethyl thiuram	
disulphide	
Paraffin Wax	

2.5.1 Compounding of Natural Rubber

The compounding of natural rubber was carried out in accordance with ASTM D4295-89 using the two-roll mill maintained at 70^{0}C to avoid cross-linking during mixing. The process involves preheating the rolls of a two-roll machine prior to the introduction of crumb rubber between the moving roll which rotate counter clockwisefor mastication to take place. After 5 minutes, the additives were added for effective compounding. Sulphur was added last to introduce 3-dimensional network structure into the rubber. The sequence is presented in Table 2.

a b
c d

Plate II: Two-roll Mill (b), Hydraulic Press (b), Rubber Compounding (c) and Compounded Rubber (d)

Table 2: Mixing Steps and Mixing Time in Rubber Compounding

Mixing Step	Time (min)
Rubber Mastication	3.00
Addition of Stearic	1.00
Acid	1.00
Addition of Zinc Oxide	5.00
Addition of Filler	1.00
(Buffing)	1.00
Addition of MBTS	1.00
Addition of TMTD	2.00
Addition of Paraffin	

Wax Addition of Sulphur	

Total	**15.00**

2.5.2 Cure Characteristics of Compounded Rubber

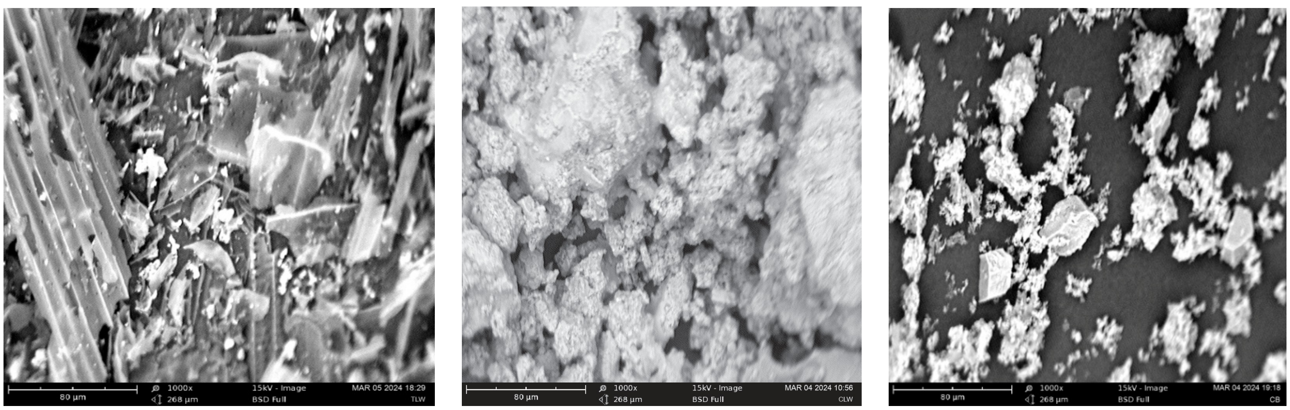
The cure characteristic was determined using Mosanto rheometer in accordance with ASTM D1415-88. The process involved cutting about 10g of the compounded unvulcanized rubber sample and placing it directly on the disc, which operates electronically. The cut test piece is then allowed to stay on the disc for about 30min. The cure time, cure temperature, scorch time, minimum and maximum modulus results of the sample were determined on the rheograph. These rheograph results were used to vulcanize the rubber samples using a hydraulic press.

3.0 Results and Discussion

3.1 Results

The results of these research findings are presented and discussed below.

3.1.1 Filler Characteristics (SEM)



Treated leather waste (a) Carbonized leather waste (b) Carbon black (c) Plate III: Micro-Graphs of the different fillers (a – c)

Table 3: Result of Natural Rubber Characteristics

Grade	NSR -10 (%)
Plasticity Retention Index	75.50
Dirt Content	0.08
Ash Content	0.65
Volatile Matter	0.72

3.1.2 Cure Characteristics of Vulcanizates

Table 4: Cure Characteristics of Vulcanizates

Filler Loading (phr)	Scorch Time, ts_2 (Mins)	Cure Time, t_{90} (Mins)	Minimum Modulus (kg-cm)	Maximum Modulus (kg-cm)	Curing Rate Index (min^{-1})
Unfilled	3.35	6.22	4.96	14.32	1.25
10	(2.59) [2.05] {1.78}	(6.04) [5.93] {4.77}	(4.10) [4.76] {5.00}	(16.70) [19.44] {20.04}	(1.27) [1.31] {1.40}
20	(2.27) [2.04] {1.65}	(5.81) [4.85] {4.79}	(4.72) [4.84] {5.17}	(19.97) [22.66] {23.53}	(1.31) [1.36] {1.45}
30	(2.15)	(5.70)	(4.80)	(22.37)	(1.37)

	[1.88] {1.59}	[4.73] {4.66}	[5.01] {5.19}	[23.17] {24.42}	[1.44] {1.52}
40	(2.08) [1.68] {1.27}	(5.64) [4.65] {4.57}	(4.88) [5.15] {5.75}	(22.41) [23.55] {24.82}	(1.45) [1.51] {1.60}
50	(2.02) [1.61] {1.19}	(5.56) [4.51] {4.45}	(5.08) [5.53] {5.97}	(25.08) [26.19] {27.95}	(1.53) [1.65] {1.74}

Key: Treated Leather Waste Filler Compounded Rubber (TLW)
Carbonized Leather Waste Filler Compounded Rubber [CLW]
Carbon Black Compounded Rubber{CB}

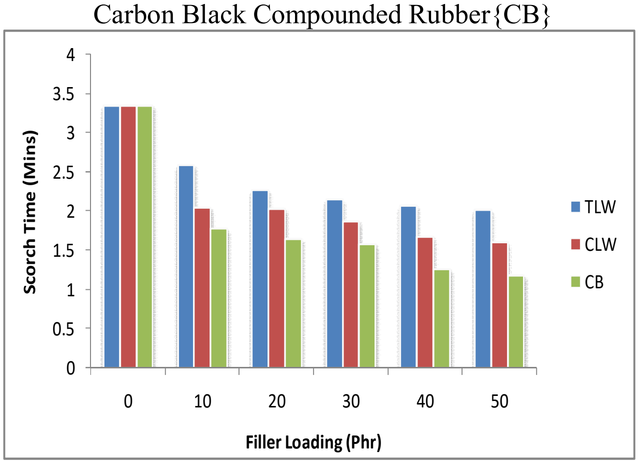


Figure 1: Scorch Time of filled Vulcanizates

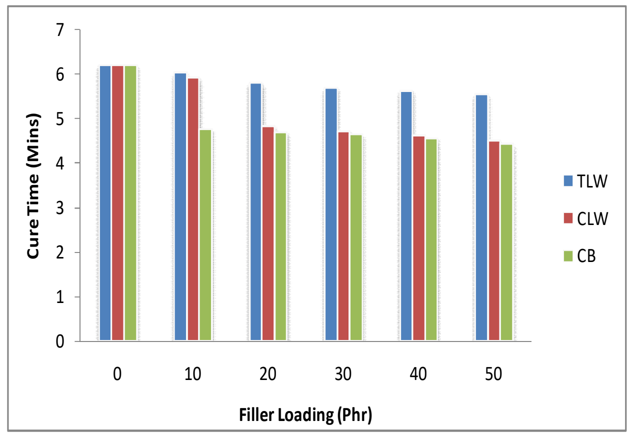


Figure 2: Cure Time of filled Vulcanizates

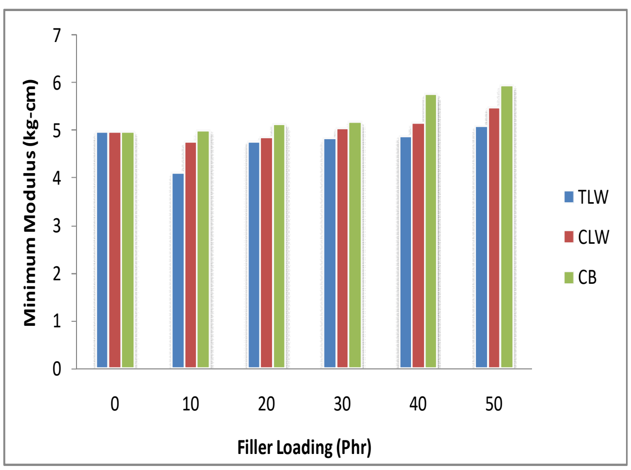


Figure 3: Minimum Modulus of filled Vulcanizates

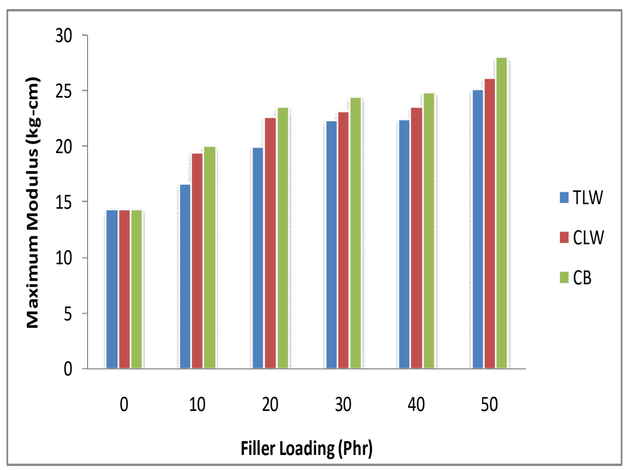


Figure 4: Maximum Modulus of filled Vulcanizates

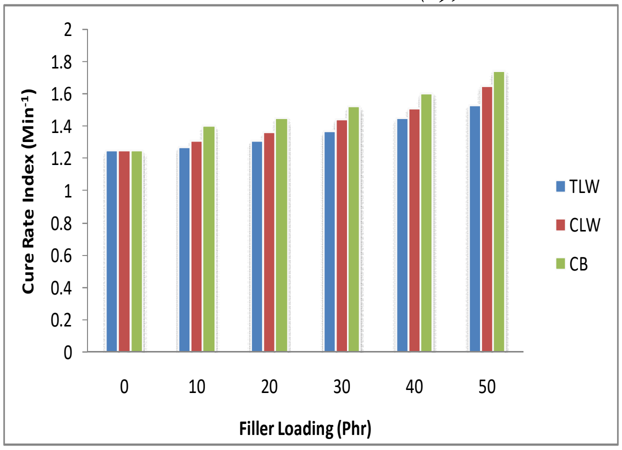


Figure 5: Cure Rate Index of filled Vulcanizates

3.2 Discussion

3.2.1 Results of Filler Characteristics

SEM micrographs in plate III (a –c) showed the morphology of the various fillers of treated leather waste (TLW), carbonized leather waste (CLW) and carbob black (CB) respectively. The progressive interphase at the transition zone of both fillers was showed. The results were interpreted in reference to ASTM C1723 (2010). The SEM morphology of the TLW powder showed a dense structure with the production of within the structure (Musa et al., 2017). However, the presence of highly define packed particles was present in the micro structure of the CLW revealing minimum pores and voids which showed that CLW will aid the interaction of filler particles in matrices and propagate adherence when incorporated in polymer matrices (Cardona et al., 2017). The micro structure of the CLW fillers showed aggregations in the packing order within the surface structure revealing tendencies to fracture surfaces which may reduce compaction when incorporated into polymer matrices. However, due to the large surface area available to network and interact with matrices, there are high tendencies that both fillers will adhere adequately with the matrices (Patnaik et al., 2010). Large surface area improves compaction and filler incorporation in polymer matrices when compared to small surface area networks (Omran et al., (2010). Stress may initiate the propagation of micro-cracks in the fillers, which further decreased the strength of vulcanizates.

3.2.2 Results of Cure Characteristics of Vulcanizates

The rheometric characteristics of the different natural rubber (NR) vulcanizates are shown in Table where minimum torgue ($M_{L)}$, maximum torgue ($M_{H)}$, scorch time ($ts_{2)}$ and cure time (ts_{90}) respectively. The minimum torque gives an indication of the filler content in the rubber while the maximum torque in the rheograph is a measure of the cross-linking density and stiffness in the rubber matrices (Kueseng et al., 2013). In general, for all the mixes, the torque initially decreases and finally leveled-off. The initial decrease in torque was due to the softening of the rubber matrix while the increase is due to the cross-linking of the rubber (Agayev et al., 2019). The leveling- off is an indication of complete curing. It is found that the presence of fibres increases the maximum torque (Kim et al., 2015). It has been revealed that upon filler modification maximum torque increases indicating greater cross-linking due to better adhesion between the fillers and the matrices. Maximum torque value exhibited by CLW filled vulcanizates indicates maximum cross-linking. The results showed that each vulcanizate exhibited different cure characteristics due to different properties of the filler. The rough measurement of the cross-linking degree of rubber during vulcanization could be used as an indirect indication of the cross-link density of the rubber compound (Granda et al., 2016). Composites at different loadings of CLW and TLW caused a significant decrease compared with pure natural rubber. It was revealed that CLW increased curing process of natural rubber. The increment in the modulus with CB content was due to the presence of more fibrous structure which imparted more restriction to the deformation and consequently increased the vulcanizates stiffness. The larger the surface area is, the greater the interaction between the filler and rubber matrices (Zhang et al., 2010).

The structure of the CLW particles contains fibres which promoted particle dispersion in the rubber matrices. The increased dispersion lead to increased interfacial interaction and cross-linking density of both matrices The NR/CLW showed the stronger interactions between particles and natural rubber matrix. It could be found that the scorch time and optimum cure time of vulcanizates decreased with increasing filler loading (Wang et al., 2014). At a similar filler loading, TLW exhibited longer cure time when compared to CLW and CB. One reason was that the high specific surface areas and the charges on the lattice allowed the TLW to adsorb the vulcanizing and curing agents easily. The adsorption increased as the CLW content varied from 10 phr to 50 phr. The TLW contained a large number of bound water and hydroxyl groups on the surface of which could adsorb the curatives and also caused delay of the rubber compound (Zhang et al., 2019). The scorch time and optimum cure time of TLW filled were increased possibly which absorbs the curative agents and thus reduce the active sulphurating agent.

4.0 Conclusion

This study revealed the potentials of leather industry generated buffing. The obtained data clearly reveals that leather waste (buffing) as filler materials affects the finished product final properties. Natural rubber vulcanizates filled with modified leather waste (buffing) were successfully prepared at different loadings using two-roll mill and introduced as potential value added product to the industrial community for the production of floor-mats. A comparative study performed on both treated leather waste (TLW), carbonized leather waste (CLW) and carbon black (CB) filled vulcanizates indicated improved adhesion of both CLW and TLW with natural rubber resulting in the outstanding cure characteristics properties which was revealed in the results obtained. The scorch and cure time of the composites decreased with increasing filler contents while the minimum and maximum modulus as well as cure rate index increased with increasing filler contents. However, CLW and CB filled matrices had better performance when compared to TLW filled NR compounds, which consumes more of the waste material.

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IMPACT OF POLYMER WASTE AND CLIMATE MITIGATION; A REVIEW

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Abstract
The U.S. Environmental Protection Agency (EPA) defines municipal solid waste, also referred to as solid waste, as consisting of everyday items use and throw away, such as product packaging, grass clippings, furniture, clothing bottles, food scraps, newspapers, appliances, paint, and batteries. Global waste sector emissions have grown steadily and are expected to increase in the forthcoming decades especially in developing countries because of the increases in population and GDP. It is estimated that there is a stock of 86 million tons of plastic marine debris in the worldwide ocean as of the end of 2023, with an assumption that 1.4% of global plastics produced from 1950 to 2023 has entered the ocean and has accumulated there. Some researchers suggest that by 2050 there could be more plastic than fish in the oceans by weight. In some areas there have been significant efforts to reduce the prominence of free range plastic pollution, through reducing plastic consumption, litter clean up, and promoting plastic recycling. As of 2023, the global mass of produced plastic exceeds the biomass of all land and marine animals combined. This large amount of plastic waste enters the environment, with studies suggesting that the bodies of 90% of seabirds contain plastic debris

Key: Scraps, Batteries, Stock, Debris, Emission

Polymer Wastes

Polymer wastes are unwanted or unusable materials or substances which are discarded after primary use, or worthless, defective and of no useful importance (Hammer et al., 2012). According to EPA, municipal solid waste, more commonly known as trash or garbage, is typically generated in homes, schools, hospitals, and businesses. However, due to the fact that many tribal lands are characterized as rural, remote or containing expansive boundaries, illegal dumping, burn barrels and other improper solid waste disposal continue to be a challenge. The most common Polymer waste known are plastics waste pollutants. The estimation of the past, current and future emissions as well as the mitigation potential in the waste sector has many uncertainties, the most important relating to the poor quality of activity data needed for estimation of the emissions (Jambeck et al., 2015).

Figure 1: Common Plastic Products

Plastic pollution is the accumulation of plastic objects and particles (e.g. plastic bottles, bags and microbeads) in the Earth's environment that adversely affects wildlife, wildlife habitat and humans (Jang et al., 2015). Plastics that act as pollutants are categorized by size into micro, meso or macro debris. Plastics are inexpensive and durable making them very adaptable for different uses, as a result human produce a lot of plastic. However, the chemical structure of most plastics renders them resistant to many natural processes of degradation and as a result they are slow to degrade. These two factors allow large volumes of plastic to enter the environment as mismanaged waste and for it to persist in the ecosystem (Jambeck et al., 2015). Living organisms, particularly marine animals, can be harmed either by mechanical effects such as entanglement in plastic objects, problems related to ingestion of plastic waste or through exposure to chemicals within plastics that interfere with their physiology. Degraded plastic waste can directly affect humans through both direct consumption (i.e. in tap water), indirect consumption (by eating animals) and disruption of various hormonal mechanisms (Hammer et al., 2012). In 2019, 368 million tonnes of plastic are produced each year; 51% in Asia, where China is the world's largest producer. From the 1950s up to 2018, an estimated 6.3 billion tonnes of plastic have been produced worldwide, of which an estimated 9% has been recycled and another 12% has been incinerated [4].

a b c d

Figure 2: Some Global Problems of Plastic Wastes in the Environment (a to d)

Types and Causes of Plastic Wastes

There are three major forms of plastic that contribute to plastic pollution which include micro, macro, and mega-plastics. Mega- and micro plastics have accumulated in highest densities in the Northern Hemisphere, concentrated around urban centres and water fronts. Both mega and macro-plastics are found in packaging, footwear, and other domestic items that have been washed off of ships or discarded in landfills (Walker et al., 2018). Plastic debris are categorized as either primary or secondary. Primary plastics are in their original form when collected. Examples of these would be bottle caps, cigarette butts, and microbeads whereas secondary plastics, on the other hand, account for smaller plastics that have resulted from the degradation of primary plastics (Barnes et al., 2009). Almost all plastics are derived from materials (like ethylene and propylene) made from fossil fuels (mostly oil and gas). The process of extracting and transporting those fuels, then manufacturing plastic creates billions of tonnes of greenhouse gases. For example, 4% of the world's annual petroleum production is diverted to making plastic, and another 4% gets burned in the refining process. Recent studies have shown that plastics in the ocean decompose faster than was once thought, due to exposure to sun, rain, and other environmental conditions, resulting in the release of toxic chemicals such as bisphenol A. However, due to the increased volume of plastics in the ocean, decomposition has slowed down. The Marine Conservancy has predicted the decomposition rates of several plastic products. It is estimated that a foam plastic cup will take 50 years, a plastic beverage holder will take 400 years, a disposable nappy will take 450 years, and fishing line will take 600 years to degrade (Walker et al., 2018).

Major Plastic Polluter Countries

In 2018 approximate 513 million tonnes of plastics wind up in the oceans every year out of which the 83,1% is from the following 20 countries: China is the most mismanaged plastic waste polluter leaving in the sea the 27.7% of the world total, second Indonesia with the 10.1%, third Philippines with 5.9%, fourth Vietnam with 5.8%, fifth Sri Lanka 5.0%, sixth Thailand with 3.2%, seventh Egypt with 3.0%, eighth Malaysia with 2.9%, ninth Nigeria with 2.7%, tenth Bangladesh with 2.5%, eleventh South Africa with 2.0%, twelfth India with 1.9%, thirteenth Algeria with 1.6%, fourteenth Turkey with 1.5%, fifteenth Pakistan with 1.5%, sixteenth Brazil with 1.5%, seventeenth Myanmar with 1.4%, eighteenth Morocco with 1.0%, nineteenth North Korea with 1.0%, twentieth United States with 0.9%. The rest of world's countries combined wind up the 16.9% of the mismanaged plastic waste in the oceans, according to a study published by Jambeck et al (2015). Around 275 million tonnes of plastic waste is generated each year around the world; between 4.8 million and 12.7 million tonnes is dumped into the sea. About 60% of the plastic waste in the ocean comes from the following top 5 countries (Jang et al., 2015).

Table 1: Plastic Pollution Generation by Top 10 Countries (Barrett et al., 2020).

Position	Country	Plastic Pollution (10^3 Tonnes Per Year)
1	China	8820
2	Indonesia	3220
3	Philippines	1880
4	Vietnam	1830
5	Sri Lanka	1590
6	Thailand	1030
7	Egypt	970
8	Malaysia	940
9	Nigeria	850
10	Bangladesh	790

All the European Union countries combined would rank eighteenth on the list.

However, the poorly-regulated incineration in those developing nations posed considerable threats to human health and the environment. Globally, researchers estimate that the production and incineration of plastic will pump more than 850 million tonnes of greenhouse gases into the atmosphere and by 2050, those emissions could rise to 2.8 billion tonnes. Alarmingly, at least 8 million tonnes of discarded plastic also enters our oceans each year, and plastic pollution at sea is on course to double by 2030 (Jang et al., 2015).

Figure 3: Plastic Wastes Deposit in Water Ways

The smaller particles (known as micro-plastics) that break off and disperse are also unwittingly ingested by marine animals, including plankton, and some of the fish we eat. The full effects of this are still being studied, but the essential premise is when micro-plastics threaten plankton populations, more carbon will re-enter the waters and atmosphere. Given that our oceans have successfully absorbed 30-50% of atmospheric carbon produced since the start of the industrial era, it's easy to see just what's at stake. And this leads us back to the plastic consumption on land that is driving this mounting plastic pollution crisis. The more plastic we make, the more fossil fuels we need, the more we exacerbate climate change (Barnes et al., 2009). The only way to address the problem is to curb the production of plastic, especially of the single-use variety, and to ramp up recycling. Reducing plastic use and waste is a key component of WWF's work. It's critical to curb greenhouse gas emissions that are exacerbating climate change, and to protect our marine environments.

Effects of Polymer Wastes

i. Effects on Environment

The distribution of plastic debris is highly variable as a result of certain factors such as wind and ocean currents, coastline geography, urban areas, and trade routes. Human population in certain areas also plays a large role in this. Plastics are more likely to be found in enclosed regions such as the Caribbean. It serves as a means of distribution of organisms to remote coasts that are not their native environments. This could potentially increase the variability and dispersal of organisms in specific areas that are less biologically diverse. Plastics can also be used as vectors for chemical contaminants such as persistent organic pollutants and heavy metals (Ang et al., 2015).

ii. Effects on Land

Plastic pollution on land poses a threat to the plants and animals including humans who are based on the land. Estimates of the amount of plastic concentration on land are between four and twenty-three times that of the ocean. The amount of plastic poised on the land is greater and more concentrated than that in the water. Mismanaged plastic waste ranges from 60 percent in East Asia and Pacific to one percent in North America. The percentage of mismanaged plastic waste reaching the ocean annually and thus becoming plastic marine debris is between one third and one half the total mismanaged waste for that year. Chlorinated plastic can release harmful chemicals into the surrounding soil, which can then seep into groundwater or other surrounding water sources and also the ecosystem of the world. This can cause serious harm to the species that drink the water (Barnes et al., 2009). However, plastic tap water pollution remains under-studied, as are the links of how pollution transfers between humans, air, water, and soil.

iii. Effect on Flooding

Volunteers clearing gutters in Ilorin, Nigeria during a volunteer sanitation day. Even when there is adequate infrastructure for sanitation, plastic pollution can prevent drainage and impede sewage flow. Plastic waste can clog storm drains, and such clogging can increase flood damage, particularly in urban areas. For example, in Bangkok flood risk increases substantially because of plastic waste clogging the already overburdened sewer system (Hammer et al., 2012).

iv. Effects on Oceans and Seabirds

Marine plastic pollution is a type of marine pollution by plastics, ranging in size from large original material such as bottles and bags, down to micro-plastics formed from the fragmentation of plastic material. Marine debris is mainly discarded human rubbish which floats on, or is suspended in the ocean. About 80% of marine debris is plastic. Micro-plastics and nano-plastics result from the breakdown or photodegradation of plastic waste in surface waters, rivers or oceans (Ostle et al., 2019). It is estimated that there is a stock of 86 million tons of plastic marine debris in the worldwide ocean as of the end of 2013, assuming that 1.4% of global plastics produced from 1950 to 2013 has entered the ocean and has accumulated there. The 2017 United Nations Ocean Conference estimated that the oceans might contain more weight in plastics than fish by the year 2050. Oceans are polluted by plastic particles ranging in size from large original material such as bottles and bags, down to micro-plastics formed from the fragmentation of plastic material (Hammer et al., 2012). Aquatic life can be threatened through entanglement, suffocation, and ingestion. Fishing nets, usually made of plastic, can be left or lost in the ocean by fishermen. Known as ghost nets, these entangle fish, dolphins, sea turtles, sharks, dugongs, crocodiles, seabirds, crabs, and other creatures, restricting movement, causing starvation, laceration, infection, and, in those that need to return to the surface to breathe, suffocation (Jambeck et al., 2015).

v. Effects on Humans

Compounds that are used in manufacturing pollute the environment by releasing chemicals into the air and water. Some compounds that are used in plastics, such as phthalates, bisphenol A (BRA), polybrominated diphenyl ether (PBDE), are under close statute and might be very hurtful. Even though these compounds are unsafe, they have been used in the manufacturing of food packaging, medical devices, flooring materials, bottles, perfumes, cosmetics and much more. The large dosage of these compounds is hazardous to humans, destroying the endocrine system. BRA imitates the female's hormone called estrogen. PBD destroys and causes damage to thyroid hormones, which are vital hormone glands that play a major role in the metabolism, growth and development of the human body. Although the level of exposure to these chemicals varies depending on age and geography, most humans experience simultaneous exposure to many of these chemicals (Jang et al., 2015). Average levels of daily exposure are below the levels deemed to be unsafe, but more research needs to be done on the effects of low dose exposure on humans. A lot is unknown on how severely humans are physically affected by these chemicals. Some of the chemicals used in plastic production can cause dermatitis upon contact with human skin. In many plastics, these toxic chemicals are only used in trace amounts, but significant testing is often required to ensure that the toxic elements are contained within the plastic by inert material or polymer. Children and women during their reproduction age are at most at risk and more prone to damaging their immune as well as their reproductive system from these hormone-disrupting chemicals (Ang et al., 2015).

Waste Management Planning to Mitigate the Impact of Climate Change

Climate change is expected to produce more frequent and powerful natural disasters, which will increase the amount of disaster-related waste generated. Communities can adapt to these disasters and increase their resiliency by preparing for these disasters through pre-incident planning. Planning can expedite the removal of waste during and after an incident, which can reduce dangers of fire, personal injury and disease vectors and identify waste management opportunities and strategies. The affected are faced with challenges which include (Selke et al., 2015);

i. Larger quantities of waste resulting from the disaster.
ii. Wider variety of generated wastes at one time, including atypical wastes in greater quantities.
iii. Wider area of impact, possibly affecting more than one jurisdiction.
iv. Increased greenhouse gas (GHG) emissions from waste management activities, such as the transportation, treatment and disposal of large amounts of waste.

v. Insufficient waste management capacity to handle surges in necessary recycling, treatment and disposal.
vi. Greater chances of waste management facilities being impacted by the disaster, resulting in possible decrease to existing capacity for generated wastes and reduction of available waste management options.

Figure 4: Global Implications of Polymer Wastes on Climate Change

According to the EPA, waste prevention and recycling- jointly referred to as waste reduction are potent strategies for reducing greenhouse gases. There are many co-benefits of implementing waste management plans in Indian Country beyond greenhouse gas reduction. These co-benefits include a healthier airshed, reduced illegal dumping, and community beautification. What tribes can do: The four main tenets of waste reduction are reduced, reuse, recycle and compos. For waste that cannot be prevented, tribes can provide recycling, garbage pickup, and self-haul options as safe and healthy means to managing waste disposal. Waste reduction is a powerful tool for mitigating greenhouse gas emissions from the municipal waste sector. Integrated waste management plans (IWMPs) are an effective means of implementing important waste reduction strategies within communities (Hammer et al., 2012). EPAs Office of Resource Conservation and Recovery (ORCR) has developed a brochure to assist tribes, titled 3 Steps to Developing Tribal Integrated Waste Management Plan (IWMP). Again, in addition to greenhouse gas mitigation from waste reduction, there are many benefits to developing and implementing IWMPs. ORCR identifies several of these benefits (Thompson et al., 2009):

i. Lowers operating costs for waste management
ii. Increases program efficiency
iii. Reduces the use of open dumps
iv. Improves protection of human health & the environment

Adapting to the Waste-related Impacts of Climate Change (Walker et al., 2018)

Of the almost 3 million tonnes of plastic that Australia produces each year, 95% is discarded after a single use. Less than 12% is recycled, which leaves a staggering amount to be disposed of - in landfills or incinerated. We used to rely on countries like China, Myanmar and Cambodia to handle our waste plastic. Although energy and water reduction efforts are critical to reducing our impact on the environment, waste reduction is just as important.

a Identifying Strategies to Expedite Removal of Disaster-Related Waste During a Disaster

i. Reduces dangers of fire, personal injury and disease vectors
ii. Limits number of times waste is handled during cleanup
iii. Increases probability that waste will be separated into different waste streams, instead of co-mingled into large piles of waste, which facilitates reuse, recycling, treatment and proper disposal of different waste streams

b Reuse and Recycling Program to Enhanced Scaled up to Handle Disaster-Related Wastes

i. Maximizes reuse and recycling opportunities available to the community within and across jurisdictional lines during a disaster
ii. Maintains a robust and viable reuse and recycling infrastructure, such as recycling facilities and end markets for reused and recycled products
iii. Encourages green building programs

c Opportunities for Source Reduction and Hazard Mitigation Before Disaster Occurs

i. Decreases the total amount of waste that may be generated (e.g., by raising minimum floor/foundation elevations in low-lying areas or updating building code requirements so that more resilient building materials and strategies that increase a buildings capacity to withstand greater wind, rain or snow loads are incorporated into building design and construction)
ii. Eliminates the generation of potentially problematic wastes (e.g., retrofitting PCB transformers to reduce PCB-contaminated wastes)

Climate Mitigation

The term mitigation refers to efforts to cut or prevent the emission of greenhouse gases - limiting the magnitude of future warming. It may also encompass attempts to remove greenhouse gases from the atmosphere (Van-Minnen et al., 2008). Climate change mitigation consists of actions to limit global warming and its related effects. This involves reductions in human emissions of greenhouse gases (GHGs) as well as activities that reduce their concentration in the atmosphere. It is one of the ways to respond to climate change, along with adaptation. Mitigation of climate change may also be achieved by changes in agriculture, reforestation and forest preservation and improved waste management. Methane emissions, which have a high short-term impact, can be targeted by reductions in cattle and more generally by reducing meat consumption (Boysen et al., 2017).

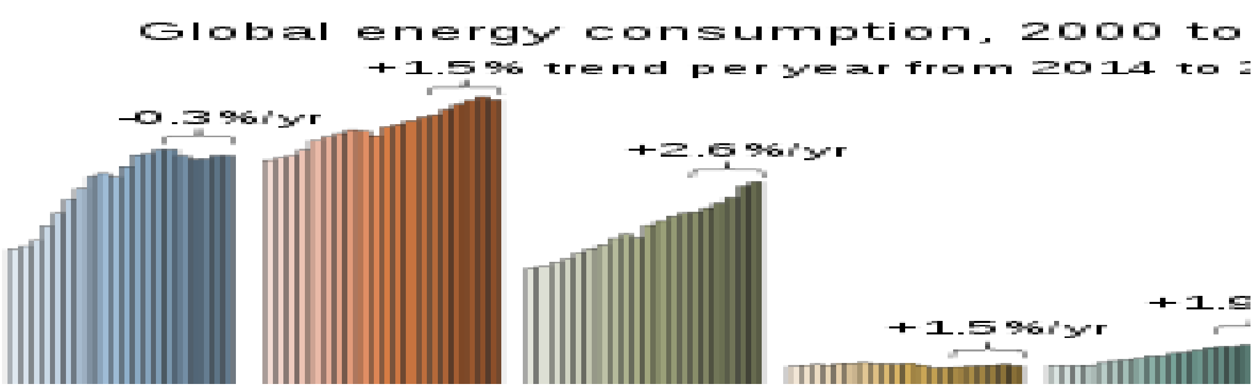

Figure 5: Global Energy Consumption

Drivers of Global Warming

GHG emissions are measured in CO_2 determined by their global warming potential (GWP), which depends on their lifetime in the atmosphere. Estimations largely depend on the ability of oceans and land sinks to absorb these gases. Short-lived climate pollutants (SLCPs) including methane, hydrofluorocarbons (HFCs), tropospheric ozone and black carbon persist in the atmosphere for a period ranging from days to 15 years as compared to carbon dioxide which can remain in the atmosphere for millennia. Reducing SLCP emissions can cut the ongoing rate of global warming by almost half and reduce the projected Arctic warming by two-thirds (Boysen et al., 2017).

i. Carbon dioxide (CO_2)

Fossil fuel: oil, gas and coal (89%) are the major driver of anthropogenic global warming with annual emissions

i. Cement production (4%) is estimated at 1.42 GtCO
ii. Land-use change (LUC) is the imbalance of deforestation and reforestation. Estimations are very uncertain at 4.5 GtCO

iii. Wildfires alone cause annual emissions of about 7 GtCO
iv. Non-energy use of fuels, carbon losses in coke ovens, and flaring in crude oil production
2 mostly by fossil fuel (72%)

ii. Methane (CH_4)

Methane has a high immediate impact with a 5-year global warming potential of up to 100. For methane, a reduction of about 30% below current emission levels would lead to a stabilization in its atmospheric concentration (Jang et al., 2015).

i. Fossil fuels (32%), again, account for most of the methane emissions including coal mining (12% of methane total), gas distribution and leakages (11%) as well as gas venting in oil production (9%).
ii. Livestock (28%) with cattle (21%) as the dominant source, followed by buffalo (3%), sheep (2%), and goats (1.5%).
iii. Human waste and wastewater (21%): When biomass waste in landfills and organic substances in domestic and industrial wastewater is decomposed by bacteria in anaerobic conditions, substantial amounts of methane are generated.
iv. Rice cultivation (10%) on flooded rice fields is another agricultural source, where anaerobic decomposition of organic material produces methane [13].

iii. Nitrous Oxide (N_2O)

N_2O has a high GWP and significant Ozone Depleting Potential (ODP). It is estimated that the global warming potential of N_2O over 100 years is 265 times greater than CO_2. For N_2O, a reduction of more than 50% would be required for a stabilization (Van-Minnen et al., 2008).

i. Most emissions (56%) by agriculture, especially meat production: cattle (droppings on pasture), fertilizers, animal manure.
ii. Combustion of fossil fuels (18%) and bio fuels.
iii. Industrial production of adipic acid and nitric acid.

iv. F-Gases

Fluorinated gases include hydrofluorocarbons (HFC), perfluorocarbons (PFC), and sulphur hexafluoride (SF6). They are used by switchgear in the power sector, semi-conductor manufacture, aluminium production and a large unknown source of SF6. Continued phase down of manufacture and use of hydrofluorocarbons (HFCs) under the Kigali Amendment to the Montreal Protocol will help reduce HFC emissions and concurrently improve the energy efficiency of appliances that use HFCs like air conditioners, freezers and other refrigeration devices (Lai et al., 2004).

v. Black carbon

Black carbon is formed through the incomplete combustion of fossil fuels, biofuel, and biomass. It is not a greenhouse gas but a climate forcing agent. Black carbon can absorb sunlight and reduce albedo when deposited on snow and ice. Indirect heating can be caused by the interaction with clouds. Black carbon stays in the atmosphere for only several days to weeks. Emissions may be mitigated by upgrading coke ovens, installing particulate filters on diesel-based engines and minimizing open burning of biomass (Pimentel et al., 2005).

Climate change in Nigeria

To mitigate the adverse effect of climate change, not only did Nigeria sign the Paris agreement to reduce emission, in its national climate pledge, the Nigerian government has promised to work towards ending gas flaring by 2030. In order to achieve this goal, the government established a Gas Flare Commercialisation Programme to encourage investment in practices that reduce gas flaring. Also, the federal government has approved a new National Forest Policy which is aimed at protecting ecosystems while enhancing social development. Effort is also being made to stimulate the adoption of climate-smart agriculture and the planting of trees (Pimentel et al., 2005). One of the outcomes of the UNFCC Copenhagen Climate Conference was the Copenhagen Accord, in which developed countries promised to provide US$30 million between 2010 and 2012 of new and additional resources. The European Commission has requested its member states to define what they understand to be additional, and researchers at the

Overseas Development Institute have found four main understandings (Howe et al., 2019):

i. Climate finance classified as aid, but additional to (over and above) the '0.7%' ODA target;
ii. Increase on previous year's Official Development Assistance (ODA) spent on climate change mitigation;
iii. Rising ODA levels that include climate change finance but where it is limited to a specified percentage; and
iv. Increase in climate finance not connected to ODAm (Anderson et al., 2012).

Links between Polymer Waste and Climate Mitigation (Prestonet al., 2011)

The earth's climate system is changing in response to increasing concentrations of greenhouse gases (GHG). GHGs are emitted through a variety of human activities including electricity production, transportation, agriculture, and industry. Reducing and recycling solid waste can help to curb the emission of greenhouse gases in some important ways (Jambeck et al., 2015).

i. Emissions from extraction of resources: Trees, minerals, oil and other basic components of materials we use must be extracted from the environment. The extraction process and resulting depletion of resources releases GHGs and disrupts the ecosystem.
ii. Transportation of goods and materials: Raw materials must be transported to processing and manufacturing facilities. Most finished goods include excessive packaging, which has several stages of extraction and transportation (Van den Berg et al., 2020).
iii. Manufacturing emissions: The production of goods and materials also has associated emissions. This can be direct emissions output from the manufacturing facility itself, or indirect output through the energy it uses.
iv. Landfill methane: As some materials degrade in the anaerobic environment of a landfill, methane is released. Methane is a potent GHG, with as high as 72 times the impact on warming the climate than CO_2. In California, landfills are one of the top five sources of GHG emissions.
v. Emissions from incineration: Even with emission controls, some GHG and other air pollutants are produced from these processes. Energy recovery capabilities at these facilities can help to negate climate impact (Van den Berg et al., 2020).
vi. Ocean pollution: Litter, particularly single-use plastic items, and polluted runoff enter our oceans, damaging its ecosystem. Healthy oceans are critical to maintaining a healthy climate, and acidification and warming ocean temperature may amplify overall warming.
vii. Destruction of forests: Trees cut down for the production of paper-based products can no longer filter our air or sequester carbon. We must preserve elements of our ecosystem that mitigate climate change.

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